Leveraging Open-Access Libraries for Feature Engineering in Material Discovery

Tiago J. Goncalves[a], Mohammad Khatamirad[b], *Edvin Fako*[a], *Raoul Naumann d'Alnoncourt*[b], *Michael Geske*[b], *Ansgar Schaefer*[a], *Stephan A. Schunk*[a,c,d], *Sonja Schimmler*[e], *Frank Rosowski*[a,b], *Sandip De*[a]

[a] *BASF SE, Group Research, Ludwigshafen, Germany*
[b] *BasCat - UniCat BASF Joint Lab, Technische Universität Berlin, Germany*
[c] *hte GmbH, Heidelberg, Germany*
[d] *Institute of Chemical Technology, Universität Leipzig, Leipzig, Germany*
[e] *Fraunhofer FOKUS – Institute for Open Communication Systems, Berlin, Germany*

Advances in machine learning (ML) and artificial intelligence (AI) are transforming material discovery. These methods significantly accelerate the exploration of large feature spaces but often struggle with small datasets and researcher biases. Additionally, model development for multi-promoter catalyst systems is challenged by complex interactions between catalyst components, often requiring expensive *ab-initio* calculations. This in turn, hampers the development of new descriptors for the design of novel materials.

In this work, we employ a data-driven approach tailored for small datasets that does not rely on prior knowledge of the studied system. Initially, an extended set of descriptors are generated through applying commutative operations to open-access atomistic properties. Additionally, interactions between catalyst components are accounted for, by introducing intrinsic promoter properties such as energies of alloy and metal oxide formation. This method is applied exemplarily to study the complex RhMn+promoter/SiO_2 catalyst system[1,2], which is tested in high-throughput experimentation for syngas to ethanol (StE) reaction.

The cross validation across multiple ML algorithms leads to a model with high accuracy. By leveraging only open-access material libraries such as materials project[3], new descriptors are obtained which go beyond the mere correlation and provide insight

into causation of observed performance trends. More importantly, the obtained model is capable of predicting new materials which were not accounted for during the training stage. Experimental studies show very good agreement with model prediction, confirming an efficient workflow for accelerated material discovery[4]. Additionally, the respective data and metadata curation is carried out according to developed standards for digital catalysis research, as outlined by NFDI4Cat[5] consortium. The proposed workflow offers the opportunity for iterative enhancement in material design (Figure.1).

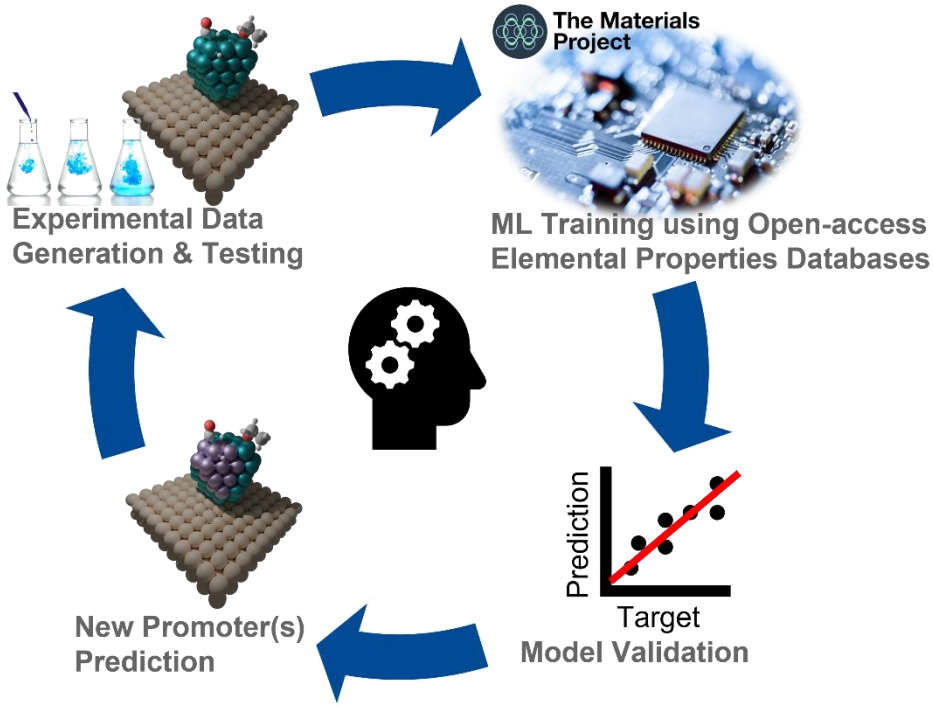

Figure 1 – Different layers of infrastructure for piloting data science tools.

References

[1] Luk, H., Mondelli, C., Ferre, D. C., et al, Chem. Soc. Rev. (2017), 46
[2] Khatamirad, M., Konrad, M., Gentzen, M., et al., MDPI Catal. (2022), 12, 1321.
[3] The Materials Project, materialsproject.org
[4] Khatamirad, Fako, E., Boscagli, C., et al, RSC Catal. Sci. Technol. (2023), 13
[5] NFDI4Cat, https://nfdi4cat.org/en/about-us/